Marcelo Soiber



Senior Backend Software Engineer | Python, APIs & Cloud

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International mobility: willing to relocate to Spain; requires employer sponsorship for a work visa

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PROFESSIONAL SUMMARY

Backend Software Engineer with 10+ years of experience in web development, payment systems, and production applications. I work with Python, PHP/Laravel, REST APIs, PostgreSQL, and AWS, contributing to the modernization of PHP/Laravel monoliths into Python microservices and the design of mission-critical APIs. I also build applied AI projects using FastAPI, RAG, LLMs, and vector search, with a strong focus on maintainability, privacy, reliability, and business needs.

PROFESSIONAL EXPERIENCE

Senior Software Developer | gsurf | Dec 2018 – Present

Santa Catarina, Brazil · Remote · Payment technology

- Provide technical leadership for the modernization and migration of PHP/Laravel monoliths to efficient Python microservices on AWS.
- Design and maintain mission-critical APIs for payment systems, prioritizing security, resilience, low latency, and operational continuity.
- Optimize relational database queries and structures, identifying and resolving application performance bottlenecks.
- Use AWS CloudFormation and cloud automation to standardize infrastructure and operational processes.
- Collaborate with stakeholders and product owners to analyze requirements, shape solutions, and translate business needs into sustainable architecture.
- Use Git-based version control and code review practices to collaborate across development teams.

PHP Web Developer | Eprocessos | Feb 2016 – Dec 2018

Garopaba, Santa Catarina, Brazil

- Developed and maintained frontend and backend features for PHP web applications.
- Designed and maintained MySQL data structures and provided technical support for applications and users.
- Contributed to the evolution of business process solutions, from requirements analysis through incident resolution.

SELECTED PROJECTS

ADR Forge | Local Architecture Decision and AI Laboratory

Technologies: Python, FastAPI, SQLAlchemy, React, TypeScript, SQLite, SSE, RAG, Transformers, PEFT, and Docker Compose.

- Designed a local-first laboratory for generating ADRs and comparing structured prompting, RAG, fine-tuning, and hybrid strategies with full traceability.
- Implemented a modular monolith, immutable revisions, context snapshots, deterministic validation, SSE monitoring, and blind human comparisons.

Credit Card Fraud Detection | Python Backend for Fraud Detection

Technologies: Python, FastAPI, REST/JSON APIs, Pydantic, pytest, PostgreSQL, scikit-learn, PyTorch, GRU/LSTM, LangGraph, and Docker Compose.

- Built a lab environment to identify anomalous transactions through sequential modeling, feature engineering, and explainable AI, with a backend organized into controllers, domain, services, repositories, and Pydantic schemas.
- Implemented APIs for prediction, asynchronous training, transaction import, PostgreSQL persistence, LLM service integration, and a monitoring dashboard to support human risk analysis.
- Created unit and integration tests with pytest. The project uses public synthetic datasets and is presented as an experimental system, not a production solution.

Knowledge Hub | Private AI-Powered Knowledge Base

Technologies: Python, FastAPI, REST/JSON, Angular, PostgreSQL, pgvector, Docker, MCP, embeddings, local models, and OpenAI-compatible models.

- Designed an application to organize and ingest documents and notes, perform semantic search, and generate answers grounded in stored content.
- Implemented a RAG architecture with a FastAPI backend, context retrieval, an Angular interface, and agent integration through an MCP server.
- Prioritized local inference, access control, data privacy, and clear separation between the API, persistence layer, and external integrations.

Knight's Tour | Interactive Genetic Algorithm Application

Technologies: TypeScript, Node.js, SQLite, and Server-Sent Events.

- Built an application to explore the Knight's Tour problem using a genetic algorithm and visualize its evolution in real time.
- Ran generations as asynchronous tasks, streamed progress through SSE, and stored execution history in SQLite.
- Implemented fitness evaluation, selection, crossover, mutation, and partial restarts when progress stagnated.

TECHNICAL SKILLS

Python & quality: Python, object-oriented programming, FastAPI, Pydantic, pytest, unit and integration testing, virtual environments, and dependency management.

Backend & integrations: Python, FastAPI, PHP, Laravel, REST APIs, JSON, microservices, system integrations, and asynchronous workflows.

Cloud & DevOps: AWS, AWS CloudFormation, Docker, Docker Compose, Git, pull request collaboration, and cloud process automation.

Data & architecture: PostgreSQL, MySQL, pgvector, SQLite, database optimization, software architecture, resilience, and performance.

AI & data: RAG, LLMs, embeddings, LangChain, LangGraph, LangSmith, MCP, scikit-learn, PyTorch, Transformers, PEFT, and model evaluation.

Web: JavaScript, TypeScript, Node.js, React, Angular, and Server-Sent Events.

Developing knowledge: CI/CD pipelines, application-level Kubernetes, and GitOps practices with Argo CD/Flux.

EDUCATION

Postgraduate Degree in Software Engineering with Applied AI | UNIPDS | 2026–2027 | In progress

Bachelor's Degree in Computer Science | Universidade do Sul de Santa Catarina (Unisul) | Final project presented in 2015

Technical Diploma in Computer Programming | CEDUP | 2009–2011

ADDITIONAL TRAINING

Advanced PHP — CEDUP · **Intermediate PHP** — CEDUP · **Android Mobile Application Development** — Unisul

Computer and Network Assembly and Maintenance — Microlins · **English** — Wizard

LANGUAGES

Portuguese: Native · **English:** Basic (A1), currently studying; able to read technical documentation · **Spanish:** Basic (A1), currently studying

AWARDS & RECOGNITION

3rd place in the Programming Marathon — 2012 and 2013 · Participant in Unisul Tech Day — 2011 and 2013

PROFESSIONAL LINKS

LinkedIn · GitHub · Portfolio · ADR Forge · Fraud Detection · Knowledge Hub · Knight's Tour